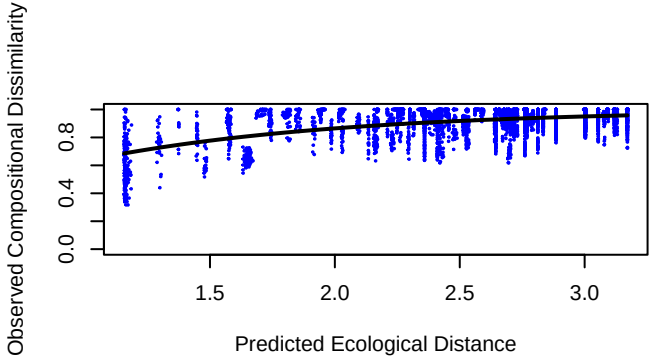
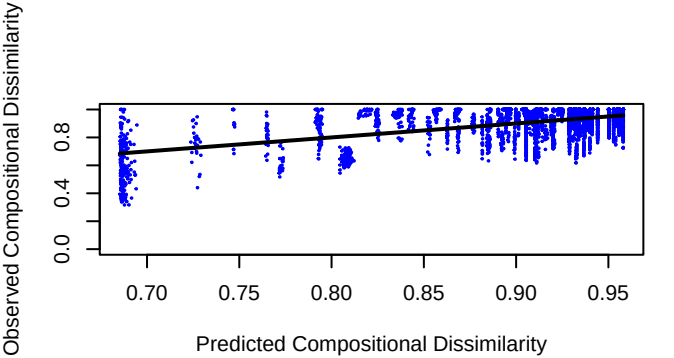
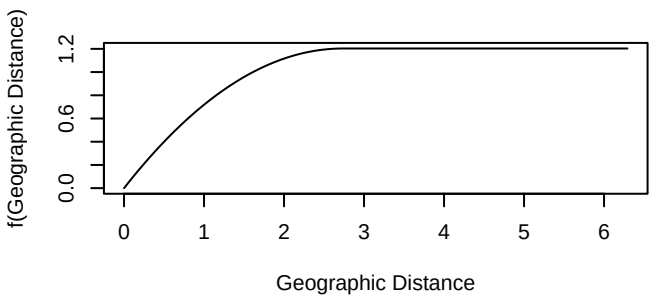
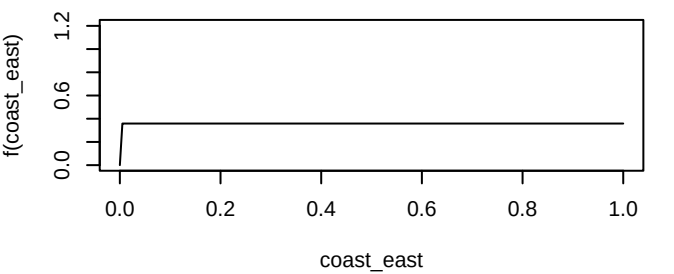
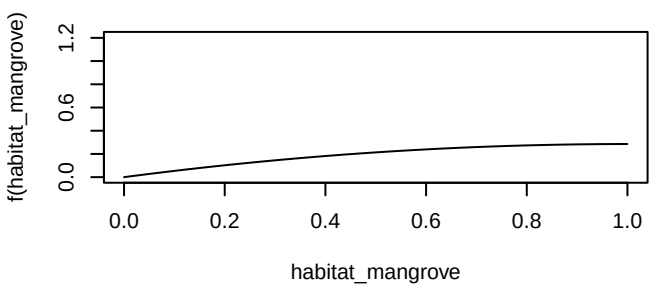
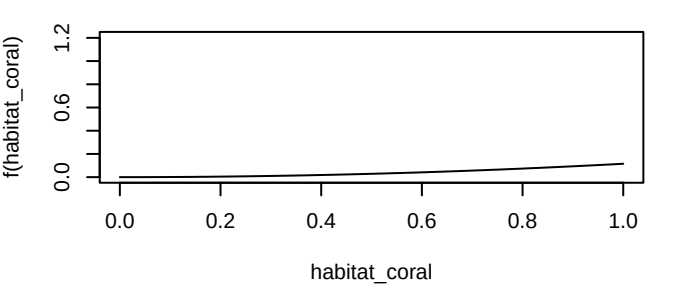

Observed Compositional Dissimilarity
0.0
0.4
0.8
1.5
2.0
2.5
3.0
Predicted Ecological Distance

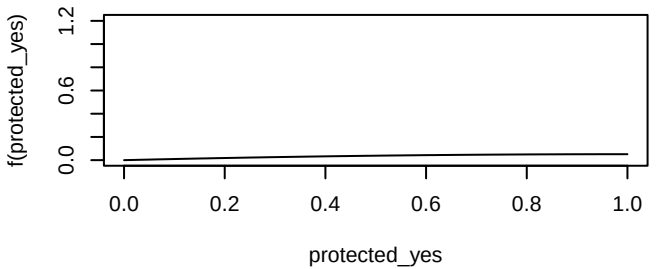

Observed Compositional Dissimilarity
0.0
0.4
0.8
0.70
0.75
0.80
0.85
0.90
0.95
Predicted Compositional Dissimilarity

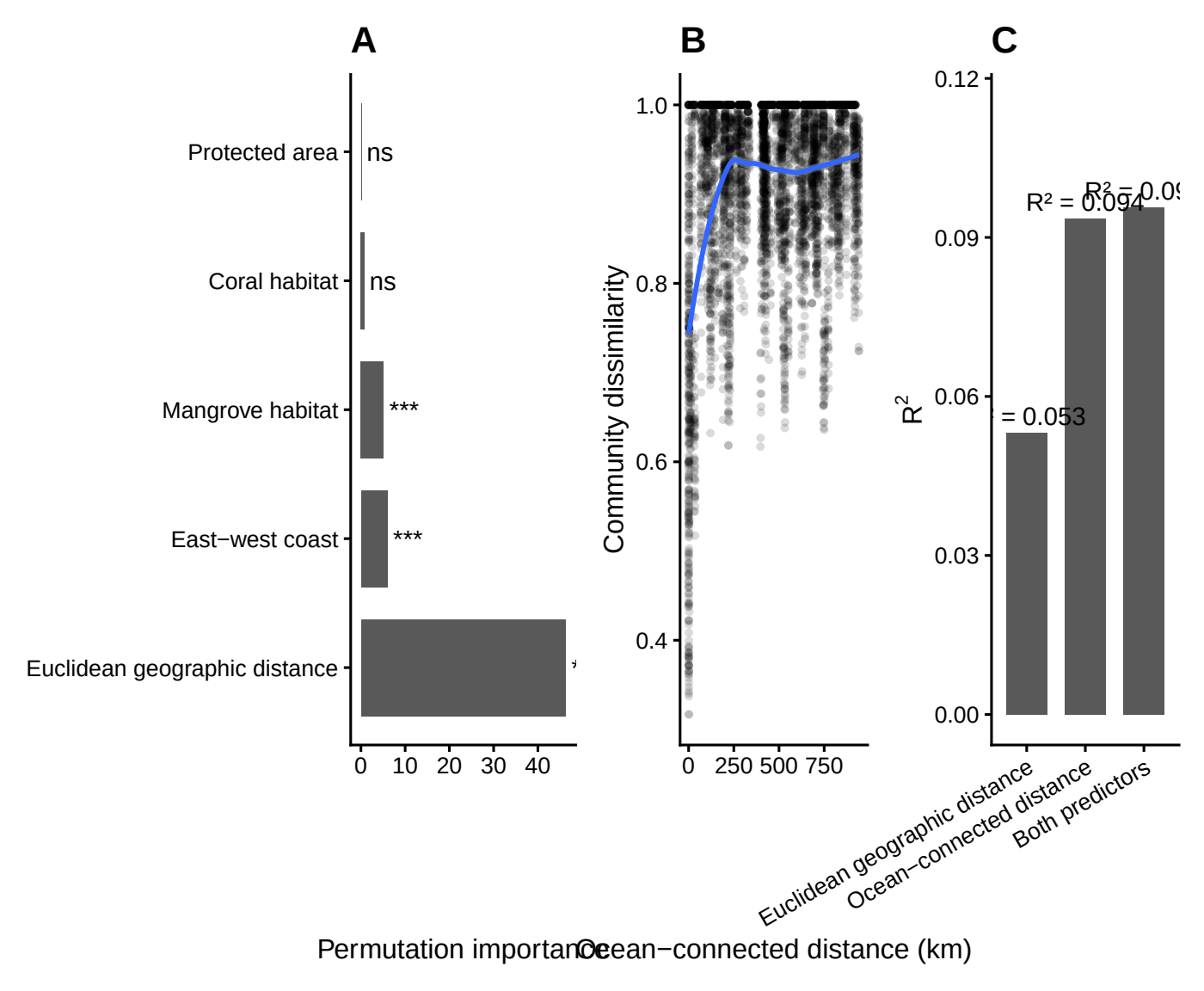

f(Geographic Distance)
0.0
0.6
1.2
0
1
2
3
4
5
6
Geographic Distance

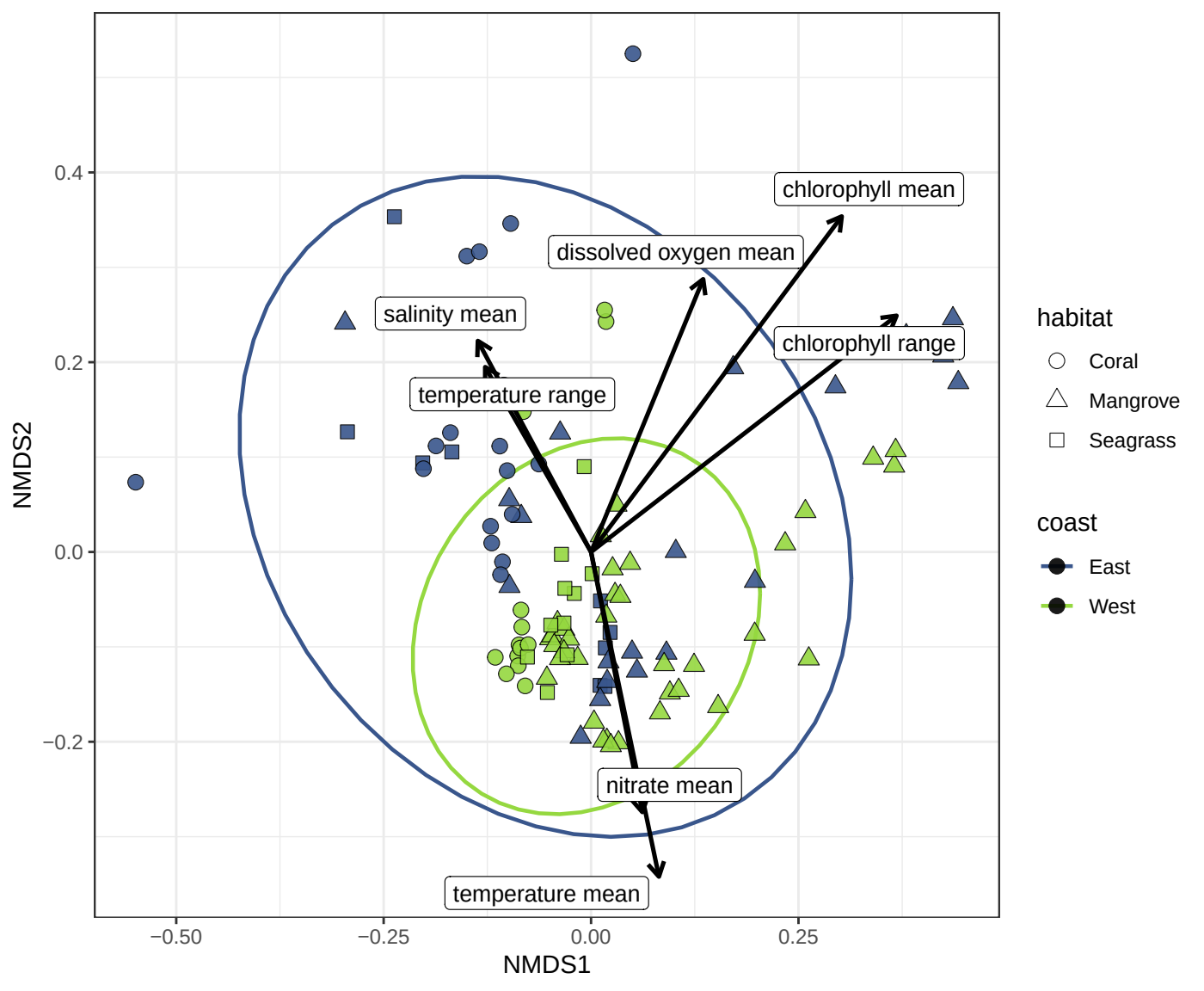

f(coast_east)
0.0
0.6
1.2
0.0
0.2
0.4
0.6
0.8
1.0
coast_east

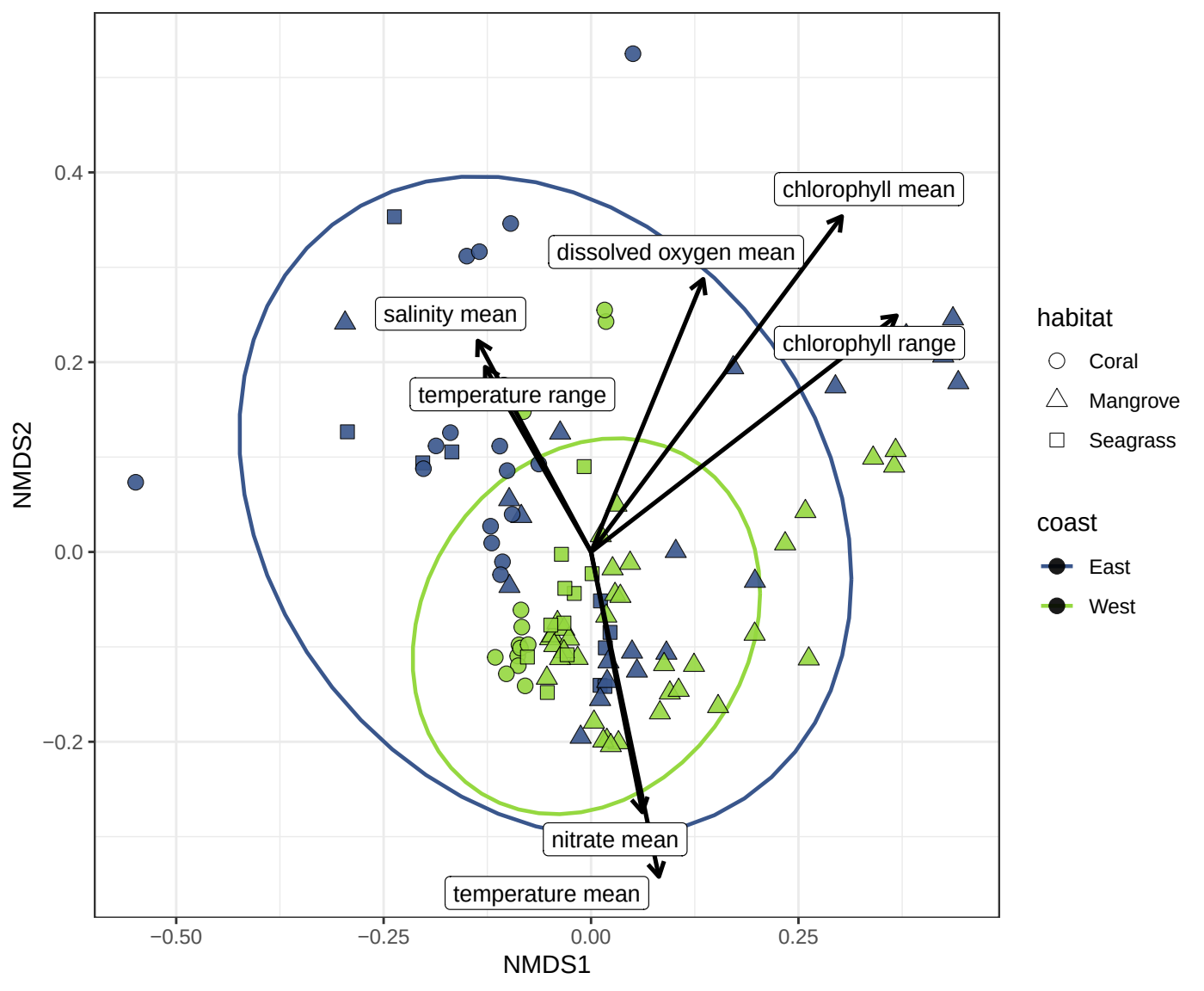

f(habitat_mangrove)
0.0
0.6
1.2
0.0
0.2
0.4
0.6
0.8
1.0
habitat_mangrove

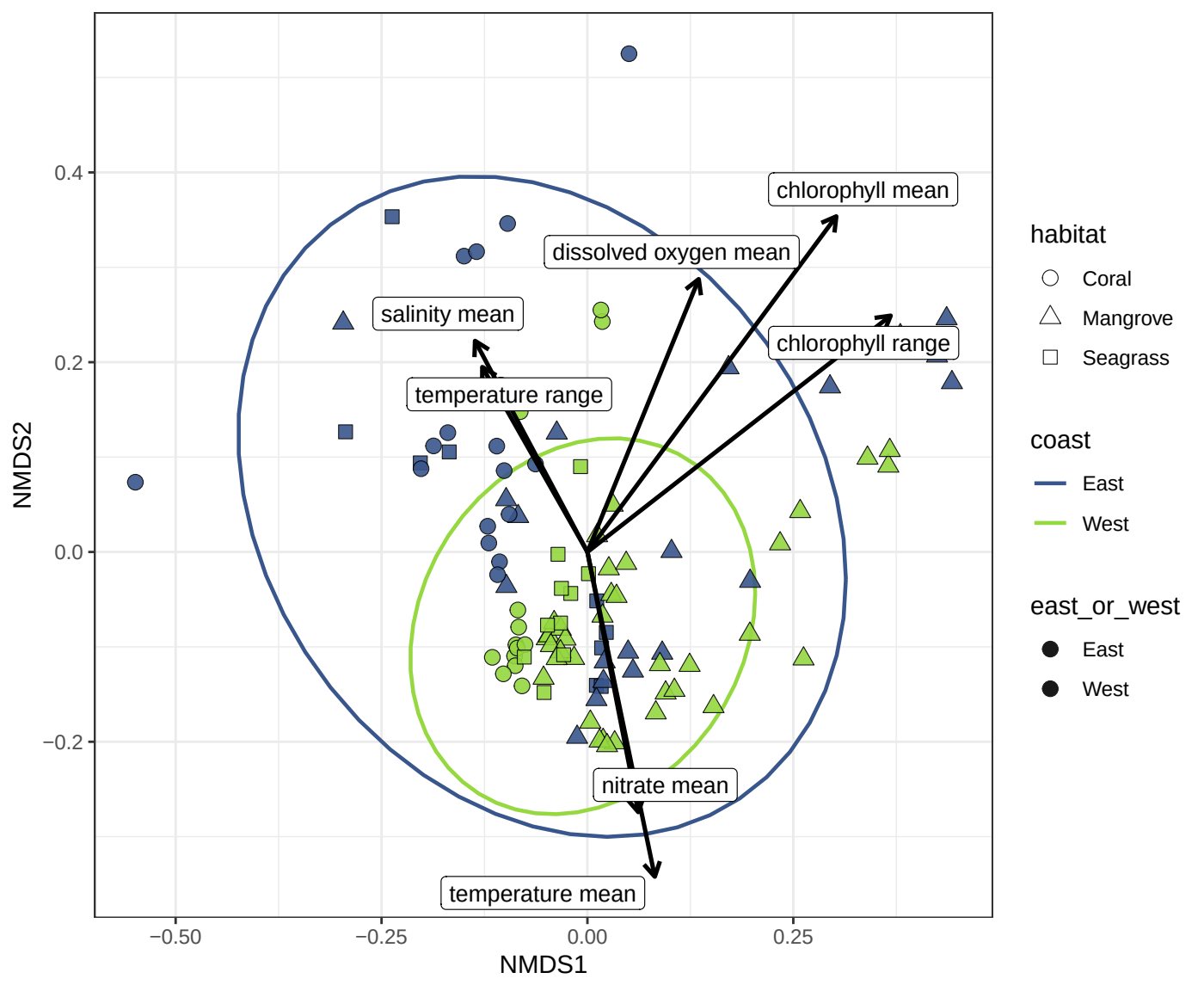

f(habitat_coral)
0.0
0.6
1.2
0.0
0.2
0.4
0.6
0.8
1.0
habitat_coral

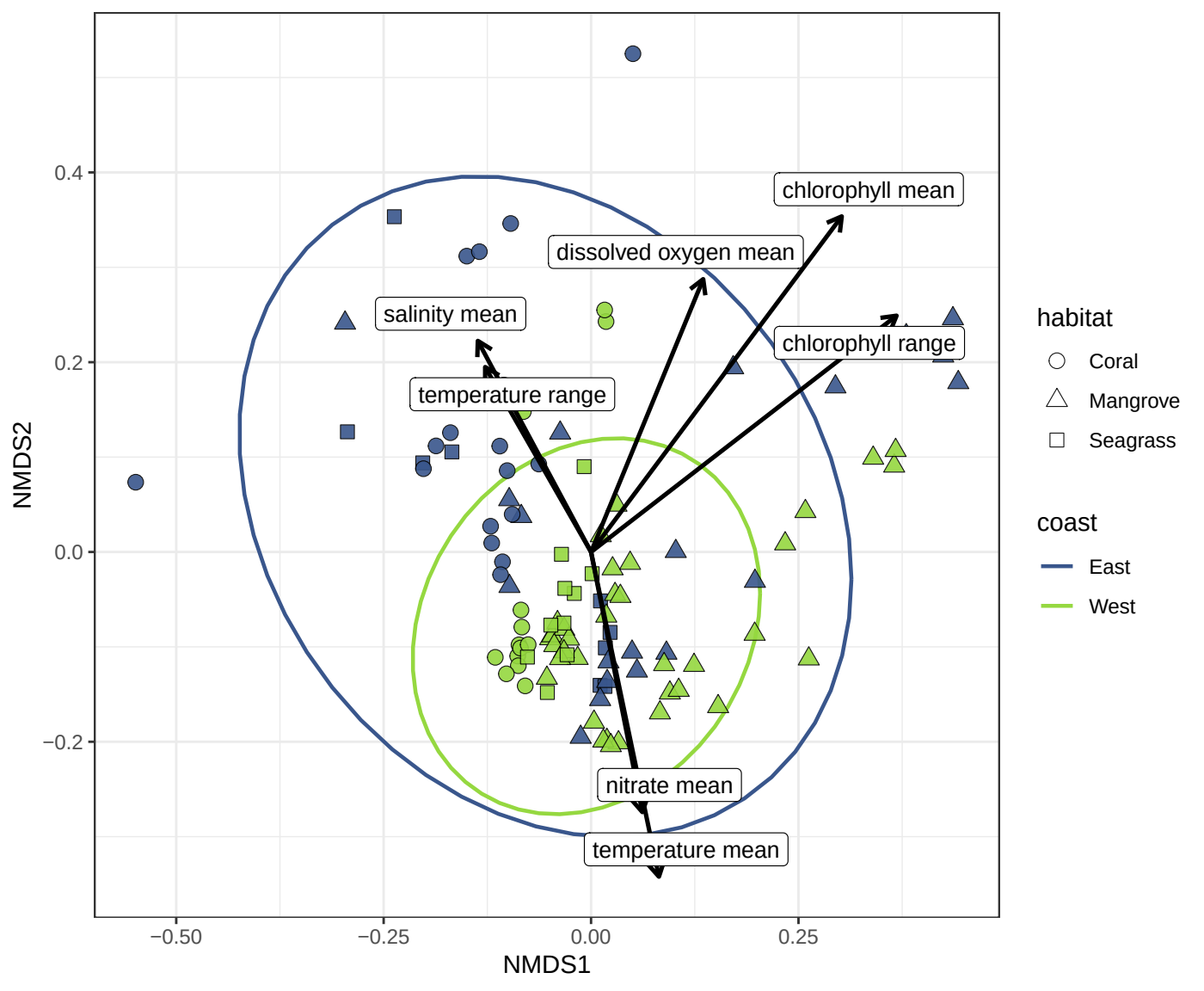

f(protected_yes)
0.0
0.6
1.2
0.0
0.2
0.4
0.6
0.8
1.0
protected_yes

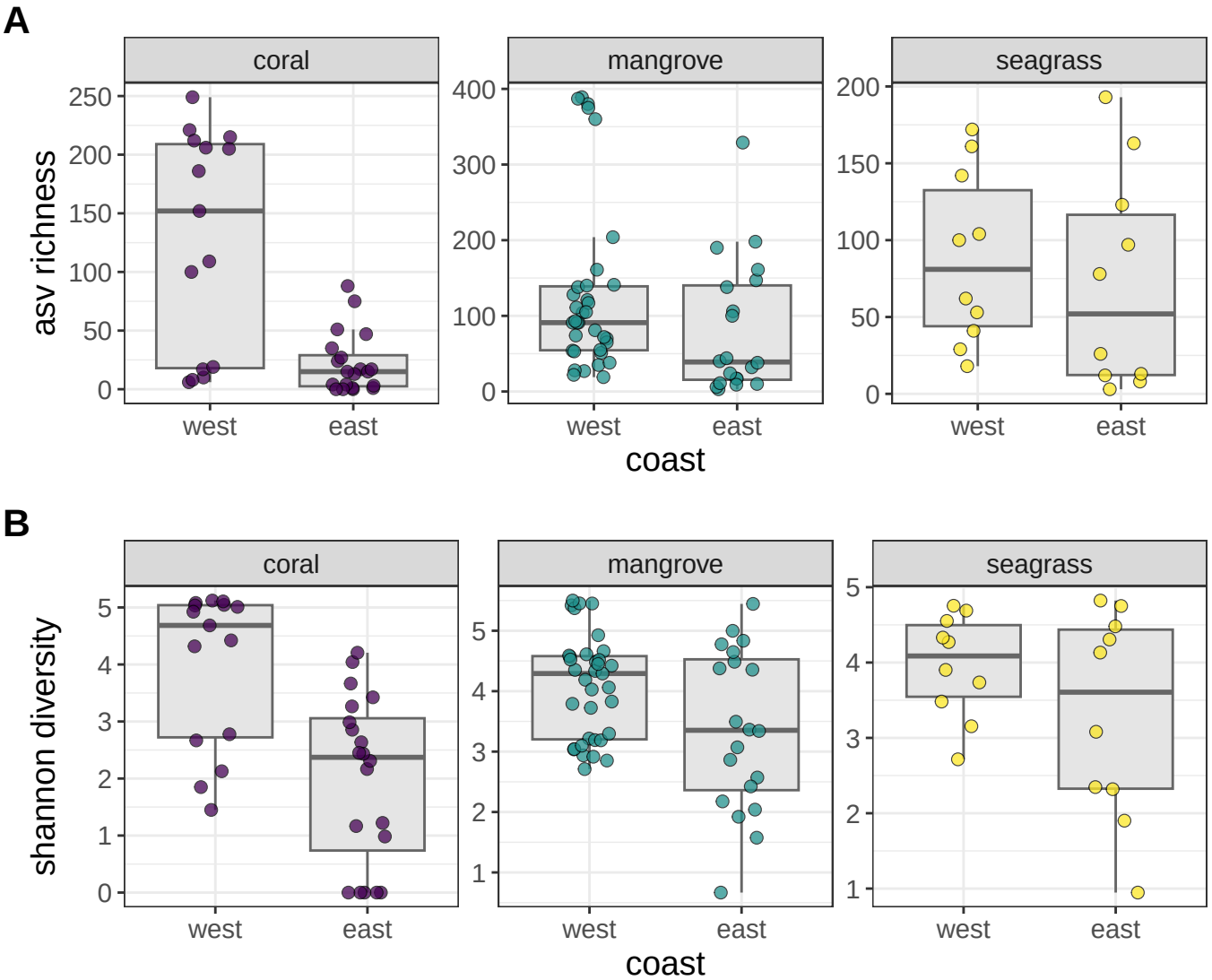

A
Protected area
ns
Coral habitat
ns
Mangrove habitat
***
East−west coast
***
Euclidean geographic distance
0
10
20
30
40
Permutation importance
B
Community dissimilarity
1.0
0.8
0.6
0.4
0
250
500
750
Ocean−connected distance (km)
C
R2
0.12
0.09
0.06
0.03
0.00
R² = 0.053
R² = 0.094
R² = 0.09
Euclidean geographic distance
Ocean−connected distance
Both predictors

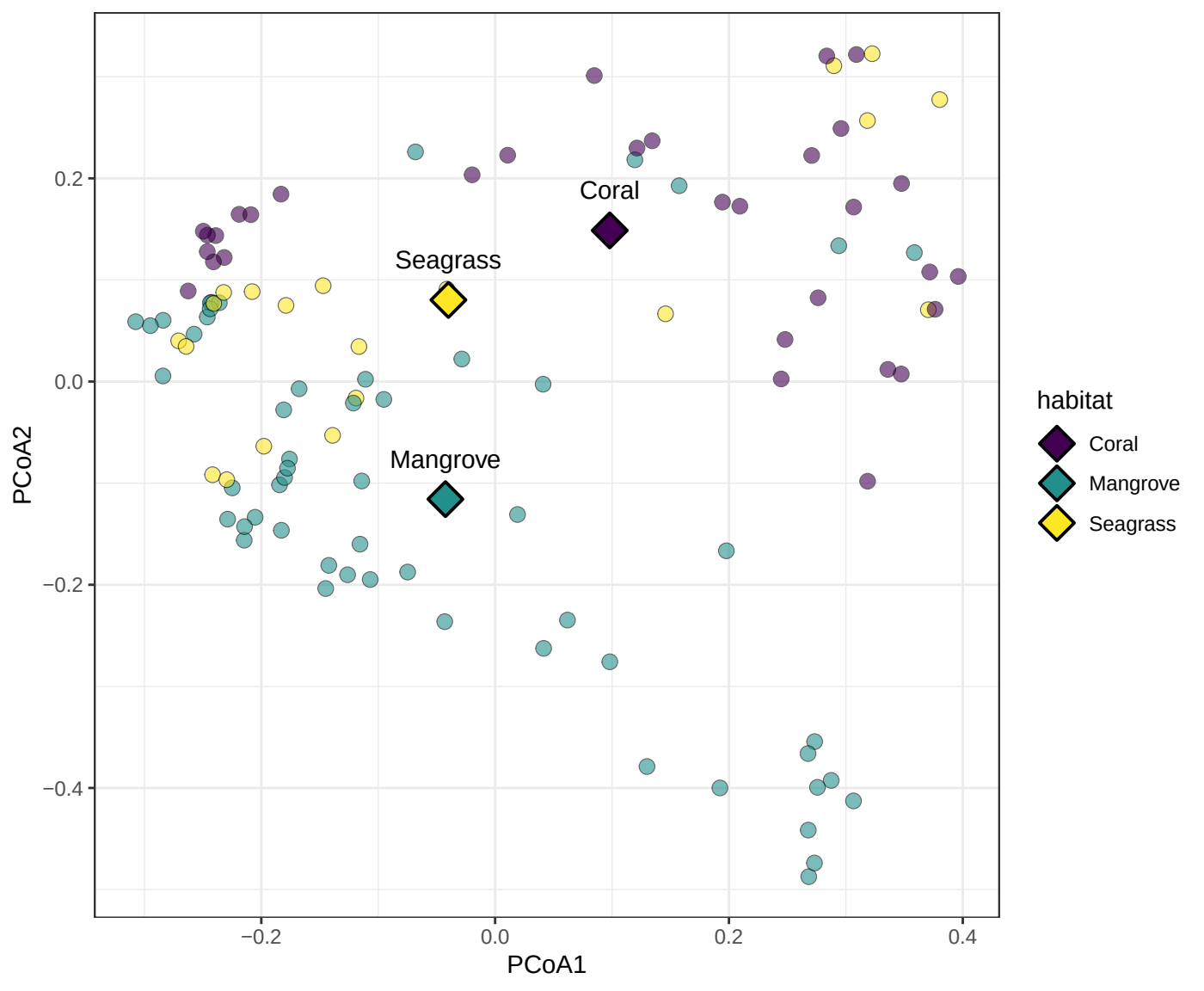

chlorophyll mean
dissolved oxygen mean
salinity mean
chlorophyll range
temperature range
nitrate mean
temperature mean
habitat
Coral
Mangrove
Seagrass
coast
East
West
NMDS1
NMDS2
0.4
0.2
0.0
−0.2
−0.50
−0.25
0.00
0.25

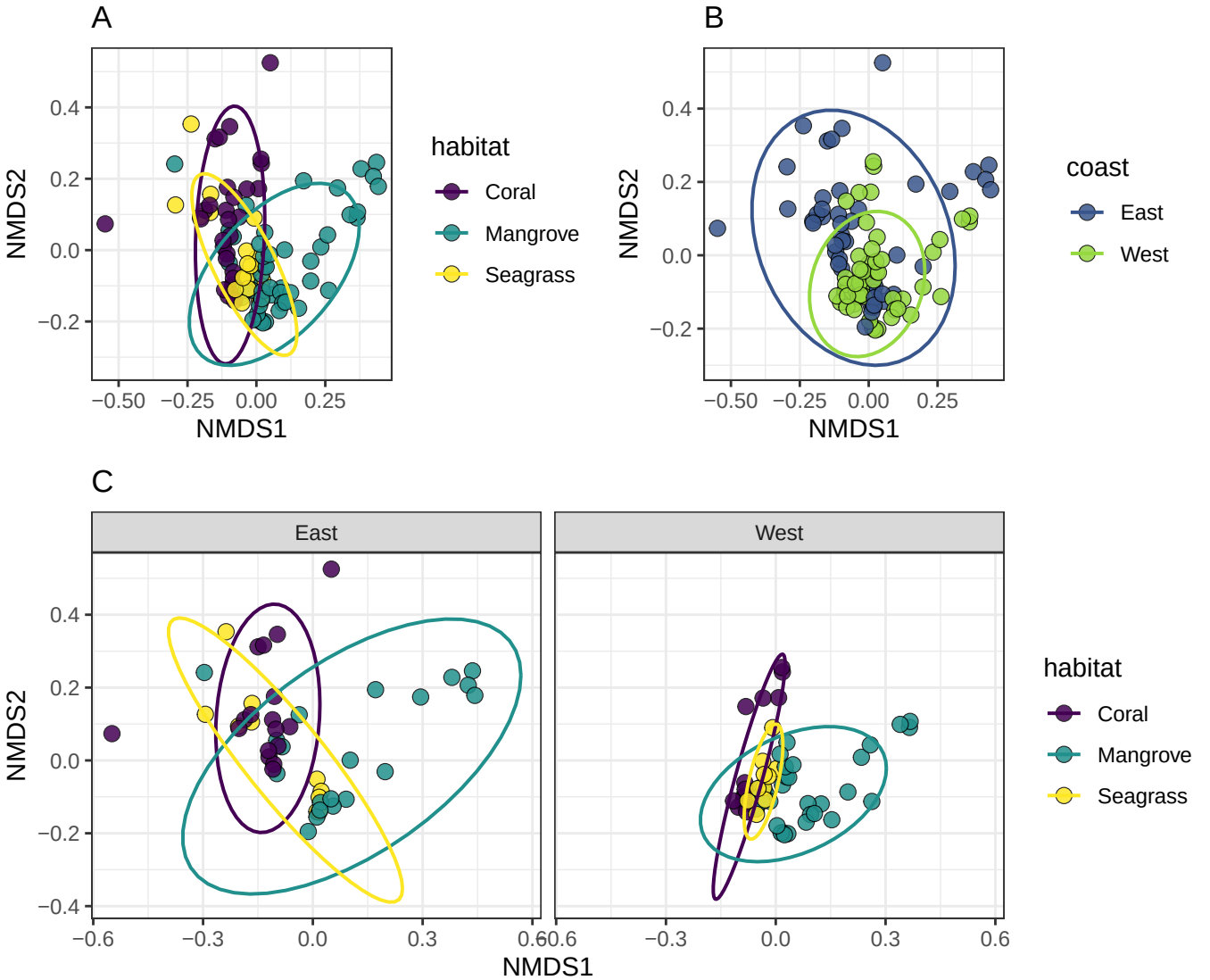

chlorophyll mean
dissolved oxygen mean
salinity mean
chlorophyll range
temperature range
nitrate mean
temperature mean
NMDS1
NMDS2
0.4
0.2
0.0
−0.2
−0.50
−0.25
0.00
0.25
habitat
Coral
Mangrove
Seagrass
coast
East
West

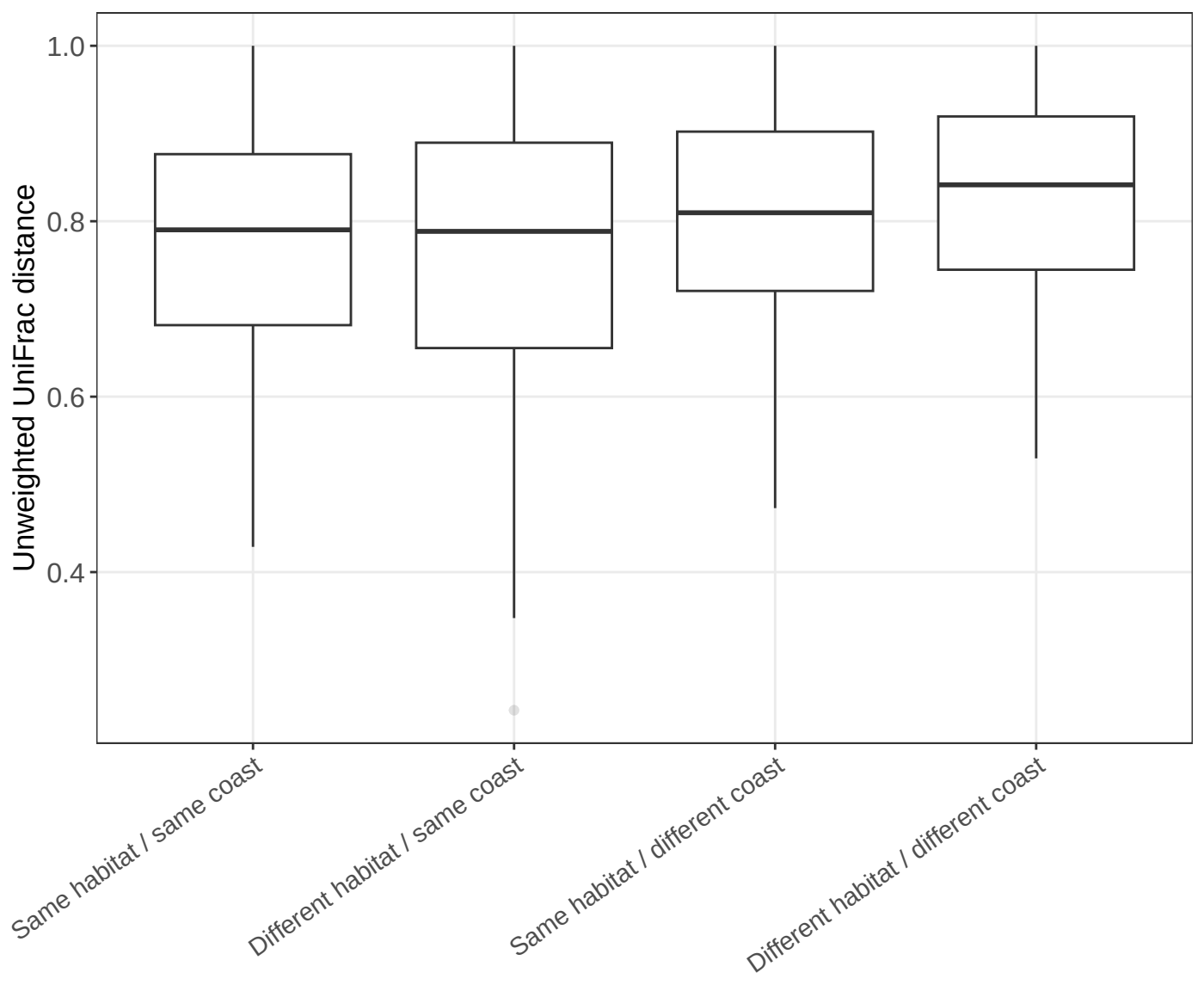

chlorophyll mean
dissolved oxygen mean
salinity mean
chlorophyll range
temperature range
nitrate mean
temperature mean
0.4
0.2
0.0
−0.2
NMDS2
−0.50
−0.25
0.00
0.25
NMDS1
habitat
Coral
Mangrove
Seagrass
coast
East
West
east_or_west
East
West


chlorophyll mean
dissolved oxygen mean
salinity mean
chlorophyll range
temperature range
nitrate mean
temperature mean
0.4
0.2
0.0
−0.2
−0.50
−0.25
0.00
0.25
NMDS1
NMDS2
habitat
Coral
Mangrove
Seagrass
coast
East
West


A
coral
mangrove
seagrass
asv richness
250
200
150
100
50
0
400
300
200
100
0
200
150
100
50
0
west
east
coast
B
coral
mangrove
seagrass
shannon diversity
5
4
3
2
1
0
west
east
coast


Coral
Seagrass
Mangrove
0.2
0.0
−0.2
−0.4
PCoA2
−0.2
0.0
0.2
0.4
PCoA1
habitat
Coral
Mangrove
Seagrass


A
B
C
NMDS1
NMDS2
habitat
Coral
Mangrove
Seagrass
coast
East
West
East
West


Unweighted UniFrac distance
1.0
0.8
0.6
0.4
Same habitat / same coast
Different habitat / same coast
Same habitat / different coast
Different habitat / different coast